2022 Minnesota Academic Standards in Mathematics Learning Progressions

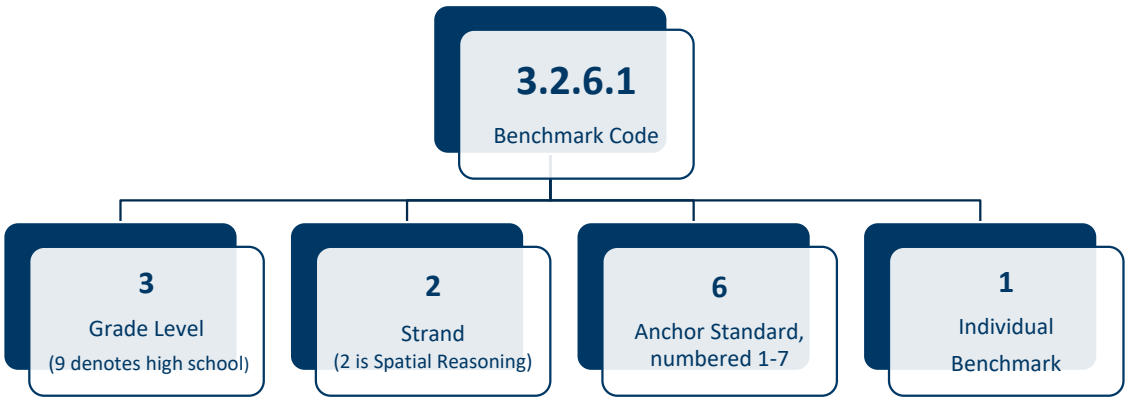
The 2022 Minnesota Academic Standards in Mathematics in the vertical format show the standards, organized by strand, with their corresponding grade-level benchmarks for grades K–High School. These learning progressions allow teachers, school leaders and administrators, and parents and caregivers to see what students should be learning in their grade level in relation to the grade levels above and below.

The 2022 Minnesota Academic Standards in Mathematics are organized into three strands: 1) Data and Probability; 2) Spatial Reasoning; and 3) Patterns and Relationships. Each strand has a series of two to three anchor standards and each anchor standard has one or more grade-level benchmarks with a 4-digit code (e.g., 0.1.1.1).

This document shows the learning progression of the Data and Probability strand from kindergarten through 12th grade, organized by anchor standards and individual benchmarks. Importantly, the standard and benchmark information below is the same as can be found in the primary Minnesota Academic Standards in Mathematics documents—available on the MDE Math implementation webpage—but is organized in a learning progression format. Please see the Spatial Reasoning and Pattern & Relationships learning progression documents, also available on the MDE Math implementation webpage.

Reading the Mathematics Standards and Benchmarks

Each benchmark has a four-digit code, which includes the grade level, strand, standard and benchmark. Each benchmark is followed by two dimensions; Mathematical Practices (MP) and Benchmark Contexts.



Dimension 1: Mathematical Practices (MP)

- MP1: Make sense of problems and persevere in solving them.
- MP2: Reason abstractly and quantitatively.
- MP3: Construct viable arguments and critique the reasoning of others.
- MP4: Model with mathematics.
- MP5: Use appropriate tools strategically.
- MP6: Attend to precision.
- MP7: Look for and make use of structure.
- MP8: Look for and express regularity in repeated reasoning.

Dimension 2: Benchmark Contexts

- ✣ Minnesota Tribal Nations Contexts
- $ Financial Literacy Contexts
- # Computer Science Contexts
- µ Modeling Contexts
- ☼ Real-world Contexts

Data and Probability Strand Anchor Standards

Strand 1: Data and Probability

"*Data are not merely numbers but numbers in context*" (Cobb and Moore, 1997).

Students will develop questions about situations impacting their everyday lives, make a plan to produce data and provide answers by organizing, describing and summarizing the data. In addition, students will develop and evaluate inferences and make predictions. They will find patterns and order from random phenomena to analyze and understand important relationships and use the concept of variability to make sense of the world.

Anchor Standard 1

Data Sciences

Identify, formulate and investigate statistical questions by collecting data considering cultural perspectives, analyzing and interpreting data and communicating results.

- A. Formulate Statistical Investigative Questions
- B. Collect and Consider Data
- C. Analyze Data
- D. Interpret and Communicate Results

Anchor Standard 2

Chance and Uncertainty

Apply and explain the concepts of probability to interpret data, generate questions, predict and make informed decisions to solve problems and communicate ideas.

Anchor Standard One: Data Sciences

Identify, formulate and investigate statistical questions by collecting data considering cultural perspectives, analyzing and interpreting data and communicating results.

A. Formulate Statistical Investigative Questions

Grade	Code	Benchmark
K▼	0.1.1.1	Notice and describe patterns in data-rich situations. (MP1, MP7) # ☼ ✣
1	1.1.1.1	Notice and describe patterns in data-rich situations **and create** statistical investigative questions with teacher guidance. (MP6, MP7) ☼ ✣
2	2.1.1.1	Notice and describe patterns in data-rich situations and create statistical investigative questions. (MP6, MP7) ☼ ✣
3	3.1.1.1	Notice and describe patterns in data-rich situations or given data sets. Ask statistical questions that can be answered with data. (MP7) ☼ ✣
4	4.1.1.1	Notice and describe patterns in data-rich situations or two given related data sets that are descriptive and comparative. Ask meaningful statistical questions that can be answered with data. (MP7) ✣
5	5.1.1.1	Notice and describe patterns in data-rich situations or given related data sets that are descriptive and comparative. Ask meaningful statistical questions that can be answered with data. (MP7) ✣
	5.1.1.2	Compare and contrast between qualitative and quantitative data. (MP3, MP6) ✣
6	6.1.1.1	Recognize a statistical question as one that anticipates variability in the data, compares differences between groups and collects categorical or numerical data related to the question, and accounts for it in the answer. (MP2) ✣
7	7.1.1.1	Formulate statistical investigative questions with data collection, keeping cultural perspectives in mind, which compare differences between groups, require a sample of a population and use categorical and/or numerical data. (MP3, MP6) $ ✣
8	8.1.1.1	Formulate statistical investigative questions, such as questions about variation, the differences between groups and associations between two numerical variables. (MP3) ✣
▲ High School	9.1.1.1	Formulate statistical investigative questions and pose hypotheses. These include questions about variation or the differences between groups, associations between quantitative and categorical variables or pairing together multiple analyses. (MP1, MP3) $ µ ☼ ✣

Anchor Standard One: Data Sciences

Identify, formulate and investigate statistical questions by collecting data considering cultural perspectives, analyzing and interpreting data and communicating results.

B. Collect and Consider Data

Grade	Code	Benchmark
K▼	0.1.1.2	Organize objects, draw pictures, or use tally marks to represent data and communicate observations. (MP3, MP6) # µ
1	1.1.1.2	Collect and use data to consider and decide what data will answer a question. Represent the data as a drawing, tally marks, frequency bar graph and digitally communicate observations. (MP1, MP3) # µ
2	2.1.1.2	Determine what counts as data to answer a statistical investigative question. Recognize that people collect data to answer questions and that data can vary. (MP1) ☼
3	3.1.1.2	Describe how data can be collected, including from surveys, grouping of items and measurement, to answer a statistical investigative question. (MP7) ☼
4	4.1.1.2	Collect and organize data to answer a statistical question, analyze variability and address missing, incomplete and bias in data. Represent data in a variety of ways, including technology. (MP1, MP5) #
5	5.1.1.3	Collect and organize data to answer statistical questions and analyze measures of center (mean and median) and variability (range). Represent data in a variety of ways, including technology. (MP5) # µ
6	6.1.1.2	Design and conduct investigations and experiments to gather data, while considering cultural perspectives, to answer statistically investigative questions considering variability and justifying choice of variables. (MP3, MP6) ✜ $ # ☼
7	7.1.1.2	Describe how statistics can be used to gain information about a population by examining a sample of the population. Explain how generalizations about a population are valid only when the sample is representative of that population and that random sampling tends to produce representative samples and support valid inferences. (MP2) ✜ ☼
8		No benchmark at this level.
▲ High School	9.1.1.2	Explain how choices concerning data collection methods can affect the quality, size, speed, accessibility and cost of the data. (MP3) $ ☼
	9.1.1.3	Analyze issues of bias by considering data collection methods and cultural perspectives. (MP3) ☼
	9.1.1.4	Explain the purposes of and differences among sample surveys, experiments and observational studies. Explain whether randomization in each allows for conclusions of causation and/or generalization of a population. (MP3) ☼

Anchor Standard One: Data Sciences

Identify, formulate and investigate statistical questions by collecting data considering cultural perspectives, analyzing and interpreting data and communicating results.

C. Analyze Data

Grade	Code	Benchmark
K▼		No benchmark at this level.
1		No benchmark at this level.
2	2.1.1.3	Collect and use data to consider and decide what data will answer a question. Represent the data as drawings, picture graphs, dot plots (a.k.a. line graphs or line plots) and with technology. Communicate observations. (MP3, MP5) # µ
3	3.1.1.3	Collect and organize data to answer a statistical question using various tools and addressing missing or incomplete data. Represent data in a variety of ways including technology. (MP1, MP5) # µ ☼
	3.1.1.5	Critically analyze data visualizations, including frequency tables, bar graphs, picture graphs or number line plots having a variety of scales to support a claim and solve situations. (MP1, MP3) $ #
4	4.1.1.4	Critically analyze data visualizations, including tables, double bar graphs, timelines, line plots or spreadsheets to support a claim and solve contextual situations. (MP3, MP7) $ # µ
5	5.1.1.4	Critically analyze data visualizations using measures of center and variability, including but not limited to double bar graphs, line graphs and line plots to support a claim and solve situations. (MP3, MP7) $ #
6	6.1.1.3	Identify, determine and interpret measures of center (mean and median) and measures of variability (range, interquartile range, mean-absolute deviation) to answer a statistically investigative question, summarizing the distribution of data using the measures of center and variability. (MP1, MP2) $ µ
	6.1.1.4	Create a visualization about a data set to describe patterns, highlight relationships or illustrate features of the distribution of the data to answer or help answer their statistically investigative question. Visualizations should represent the data in appropriate ways, including tables, dot plots, stem and-leaf plots, histograms and box plots while incorporating any other relevant information that helps to tell a story about the data. (MP5, MP6) #
7	7.1.1.1	Make inferences using statistics about population parameters based on a random sample from that population. (MP3)
7	7.1.1.4	Understand that a set of data collected to answer a statistical question has a distribution that can be described by its center, variability and overall shape. Use measures of center and measures of variability for numerical data from random samples to draw informal comparative inferences about two populations. Justify the choice of measures of center and variability, the shape of the data distribution and the context in which the data were gathered. (MP1, MP7)

Grade	Code	Benchmark
8	8.1.1.2	Construct and interpret scatter plots for bivariate measurement data to investigate patterns of association between two quantities. Describe patterns such as clustering, outliers, positive or negative association, linear association and nonlinear association. (MP5, MP8)
	8.1.1.3	Identify when to use straight lines to model relationships between two quantitative variables. For scatter plots that suggest a linear association, informally fit a straight line and assess the model fit by judging the closeness of the data points to the line. (MP5, MP6)
	8.1.1.4	Use the equation of a linear model to solve situations in the context of bivariate measurement data. Interpret the slope and intercepts in context of the variables. (MP4, MP6) $ #
▲ **High School**	9.1.1.5	Analyze and explain when arguments based on data confuse correlation and causation. (MP3)
	9.1.1.6	Compute using technology or estimate the correlation coefficient of a linear model. Interpret the linear model in the context of the data. (MP5, MP6) $
	9.1.1.7	Use the mean and standard deviation of a data set to fit it to a normal distribution and to estimate population percentages. Identify data sets for which such a procedure is not appropriate. Tools can include calculators, spreadsheets, apps or tables. (MP5, MP6) $ ☼
	9.1.1.8	Use data from a random sample to draw inferences about a population with an unknown characteristic of interest. Generate multiple samples or simulated samples of the same size to gauge the variation in estimates or predictions. (MP1, MP4) # ☼
	9.1.1.9	Use statistics appropriate to the shape of the data distribution to compare the center and spread of two or more data sets. (MP4)
	9.1.1.10	Create and analyze data displays, including scatter plots, histograms and boxplots using technology. (MP1) ☼
	9.1.1.11	Identify, create and compare statistical models with linear and exponential functions, including linear regression. Assess the reasonableness of model fit using residuals and correlation coefficients. (MP4) #

Anchor Standard One: Data Sciences

Identify, formulate and investigate statistical questions by collecting data considering cultural perspectives, analyzing and interpreting data and communicating results.

D. Interpret and Communicate Results

Grade	Code	Benchmark
K▼		No benchmark at this level.
1		No benchmark at this level.
2	2.1.1.4	Make predictions using patterns from data visualizations. (MP7, MP8) # µ
3	3.1.1.4	Make predictions and recognize that the amount and source of the data impacts the accuracy of predictions. (MP4, MP8)
4	4.1.1.3	Make predictions and recognize that how the data was collected impacts the reliability of predictions. (MP1, MP8)
5	5.1.1.5	Compare and contrast different data displays to determine how the visualizations impact analysis and interpretation. (MP3, MP8) #
6	6.1.1.5	Compare and communicate competing explanations for data trends observed, considering cultural perspectives and reasonable alternatives given the variability in findings. (MP3, MP6) ☼
7	7.1.1.5	Create a visualization about a data set, organizing and presenting the data in appropriate ways, including in tables, circle graphs and histograms, and incorporating any other relevant information that helps to tell a story about the data. (MP5, MP6) # µ
	7.1.1.6	Compare and communicate competing explanations for data trends observed, considering cultural perspectives and reasonable alternatives given the variability in findings and sampling methods. (MP3, MP6)
8	8.1.1.5	Create data visualizations about a data set. Organize and present the data in appropriate ways, including in tables and scatter plots, and incorporate other relevant information that helps to tell a story and support a claim about the data. (MP1, MP6) $ # µ ✜
	8.1.1.6	Compare and communicate competing explanations for data trends observed, considering the reasonableness of the model's predictions and correlation versus causation. (MP3, MP6) # $
▲ High School	9.1.1.12	Examine and discuss competing explanations for data trends observed such as confounding variables. Respond to competing arguments or interpretations of the data of different community groups, paying careful attention to what conclusions the data supports. (MP3, MP6) $ ☼
	9.1.1.13	Analyze and interpret data using various measures, such as difference in shapes, center and spread to draw conclusions, identify trends and describe relationships, accounting for possible effects of extreme data points (outliers). (MP1) $

Grade	Code	Benchmark
	9.1.1.14	Evaluate reports based on published data by identifying the source of the data, the design of the study and the way the data are analyzed and displayed. (MP3) $ ☼
	9.1.1.15	Identify and explain misleading uses of data along with how to use spreadsheets, tables or graphing technology to recognize and analyze distortions in data displays. Use interactive data visualizations to support and influence different points of view. (MP3) $ #

Anchor Standard 2 Chance and Uncertainty

Apply and explain the concepts of probability to interpret data, generate questions, predict and make informed decisions to solve problems and communicate ideas.

Grade	Code	Benchmark
K▼		No benchmark at this level.
1	1.1.2.1	Describe outcomes of events as impossible, possible or certain. (MP1, MP6) ☼
2	2.1.2.1	Describe the difference between possible and likely. (MP1, MP6)
3	3.1.2.1	Describe outcomes of events as impossible, certain, likely, unlikely and equally likely. (MP1, MP4)
4	4.1.2.1	Classify probability events involving dice, coins, spinners with equal and unequal partitions and blocks in a bag as impossible, certain, likely, unlikely and equally likely. (MP7)
	4.1.2.2	Use a number line to connect the values of 0 to impossible, 1/2 to equally likely, and 1 to certain. Approximate locations on the number line where likely and unlikely would occur based on the situation. (MP3, MP4)
5	5.1.2.1	List outcomes from a probability experiment in a frequency table. (MP4) ☼
	5.1.2.2	Use a frequency table to record results from an experiment to make predictions. Place predictions on a number line from 0 to 1. (MP4) # ☼
6	6.1.2.1	Determine the sample space (set of possible outcomes) for a given experiment. Sample space may be determined by the use of tree diagrams, tables or pictorial representations. (MP1) ☼
	6.1.2.2	Determine the theoretical probability of an event using the ratio between the size of the event and the size of the sample space. Represent probabilities as percentages, fractions and decimals between 0 and 1 inclusive. (MP4)
	6.1.2.3	Calculate experimental probabilities from experiments where the theoretical probability is known, recognizing that there may be differences between theoretical and experimental probability. Represent the probabilities as percentages, fractions and decimals between 0 and 1 inclusive. Use experimental probabilities to make predictions when actual probabilities are unknown. (MP4) # ☼
7	7.1.2.1	Calculate probability as a fraction of sample space or as a fraction of area. Express probabilities as fractions, decimals and percentages. (MP2)
	7.1.2.2	Approximate the probability of a chance event, where the theoretical probability is unknown, by collecting data and observing its long-run frequency. Represent the probabilities as percentages, fractions and decimals between 0 and 1 inclusive. Use approximate probabilities to make predictions when actual probabilities are unknown. (MP4) # ☼
	7.1.2.3	Recognize that, just as with simple events, the probability of a compound event is the fraction of outcomes in the sample space where the compound event occurs. (MP1)
	7.1.2.4	Represent sample spaces for compound events by decomposing the events using methods such as organized lists, tables and/or tree diagrams. For an event described in everyday language, identify the outcomes in the sample space that compose the event. (MP5) #
	7.1.2.5	Design and use a simulation within a computational tool to generate frequencies for compound events. #
	7.1.2.6	Find probabilities of compound events using organized lists, tables, tree diagrams and/or simulation via a computational tool. (MP5) # μ ☼
8		No benchmark at this level.

Grade	Code	Benchmark
▲ **High School**	9.1.2.1	Select and apply counting procedures such as the multiplication and addition principles, permutations, combinations and tree diagrams to determine the elements and size of a sample space. (MP1, MP7)
	9.1.2.2	Describe events as subsets of a sample space using characteristics of the outcomes. Use Venn diagrams, tables or lists to depict events and compute event sizes. Create representations to find unions, intersections and complements of events. (MP1) #
	9.1.2.3	Apply probability concepts, such as intersections, unions and complements of events, and conditional probability and independence, to calculate probabilities and solve situations. (MP4) # ☼
	9.1.2.4	Calculate experimental probabilities by repeatedly performing simulations or experiments involving a probability model and by using relative frequencies of outcomes. Recognize that as the number of trials increases, the closer the experimental probability approaches the theoretical probability. (MP4)
	9.1.2.5	Construct and interpret two-way frequency tables when two categories are associated with each object being classified. Use the two-way table to decide if events are independent and to calculate conditional probabilities. (MP4, MP5) ☼
	9.1.2.6	Compare the recursion to create the nth row of Pascal's triangle, the recursion to expand $(x + y)^n$, the recursion to generate all sequences of heads and tails on n coin flips and the recursion to generate all subsets of a set with n elements. Establish connections among the various representations. (MP8)
	9.1.2.7	Calculate and make decisions based on expected values using theoretical or experimental probabilities in scenarios, including finance or games of chance. (MP1, MP6) $ µ ☼
	9.1.2.8	Calculate and make decisions to obtain a desired outcome (maximum or minimum value or maximum or minimum probability) based on expected value and theoretical or experimental probabilities in scenarios such as games of chance or finance. (MP6, MP7) $ ☼